

3.5.3	Redox Equilibria	
	Redox equations	be able to apply the electron transfer model of redox, including oxidation states and half equations to d block elements
	Electrode potentials	know the IUPAC convention for writing half-equations for electrode reactions
		know and be able to use the conventional representation of cells
		understand how cells are used to measure electrode potentials by reference to the standard hydrogen electrode
		know the importance of the conditions when measuring the electrode potential, E (Nernst equation not required)
		know that standard electrode potential, $E^\ominus$, refers to conditions of 298 K, 100 kPa and 1.00 mol dm ⁻³ solution of ions
	Electrochemical series	know that standard electrode potentials can be listed as an electrochemical series
		be able to use $E^\ominus$ values to predict the direction of simple redox reactions and to calculate the e.m.f of a cell
	Electrochemical cells	appreciate that electrochemical cells can be used as a commercial source of electrical energy
		appreciate that cells can be non-rechargeable (irreversible), rechargeable and fuel cells
		be able to use given electrode data to deduce the reactions occurring in non-rechargeable and rechargeable cells and to deduce the e.m.f. of a cell
		understand the electrode reactions of a hydrogen-oxygen fuel cell and appreciate that a fuel cell does not need to be electrically recharged
		appreciate the benefits and risks to society associated with the use of these cells